Linear Model Homework 7

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0.3 Q3 Data on the relationship between corn yield (bushels per acre) and nitrogen (pounds per acre) fertilizer application were studies in Wisconsin in 1994. Use transformations to find a good model for predicting yield from nitrogen. Use a goodness of fit test to check your model. 16

0.1 Q1 The data for this question come from the US Historical Climatology network. They are the annual mean temperatures (in degrees F) in Ann Arbor going back about 150 years.

```
dat1 = read.table("aatemp.txt", header = TRUE)
diff(dat1$year)
```

```
##   [1]  1 16 10  1  1  1  1  5  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##  [26]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##  [51]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  2  1  1  3  1  1  1  1
##  [76]  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [101]  1  1  1  1  1  1  1  1  1  1  1  1  1  1
```

```
lag1.dat = dat1[9:68,]
diff(lag1.dat$year)
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

i. Is there a linear trend?

First plot temperature against year.

```
plot(temp~year, ylab = "temperature", data = dat1)
```

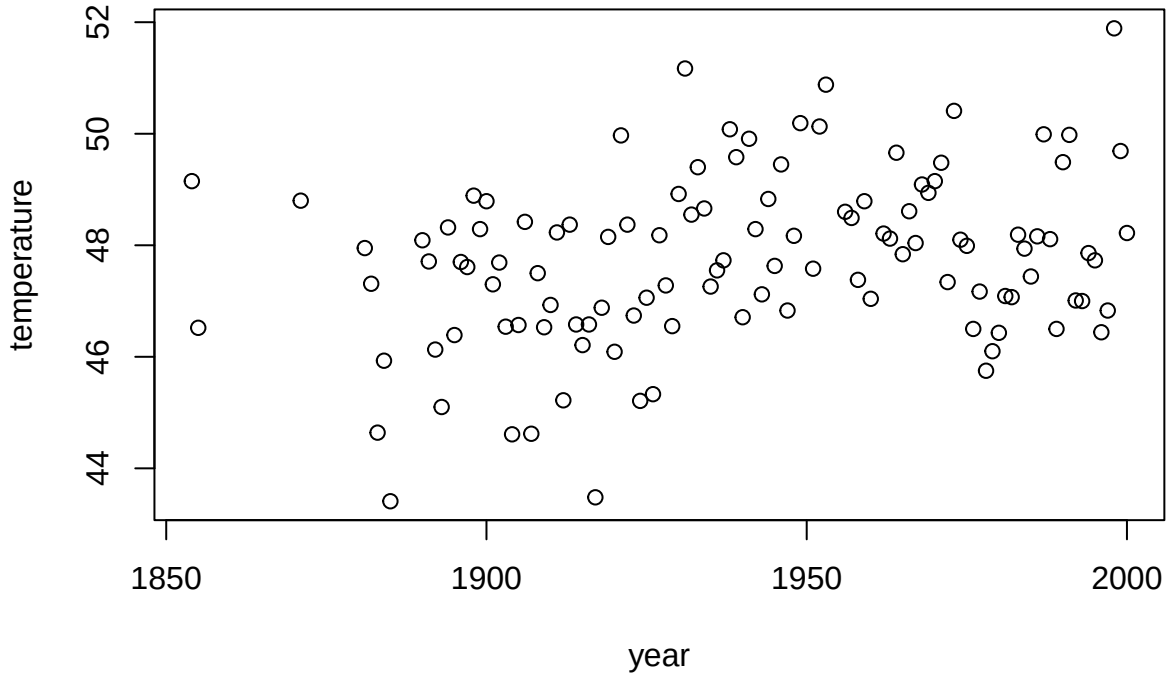


Figure 1: Scatterplot for temperature vs year

Fit the model as follows to check if there is linear trend. For convenience, let t denote year.

$$temp_t = \beta_0 + \beta_1 t + \epsilon_t.$$

The fitted result is shown in Table 1.

```
fit1 = lm(temp~year, data = dat1)
summ.kable(summary(fit1)$coef)
```

Table 1: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	24.0055103	7.3107806	3.283577	0.0013647
year	0.0122373	0.0037683	3.247430	0.0015333

Since the coefficient of t is positive and significant at 0.05 significance level, I conclude that there might be linear trend. That is, the temperature seemed to increase along the years. However, it is still possible that the trend-like pattern is present due to the correlation among the successive years.

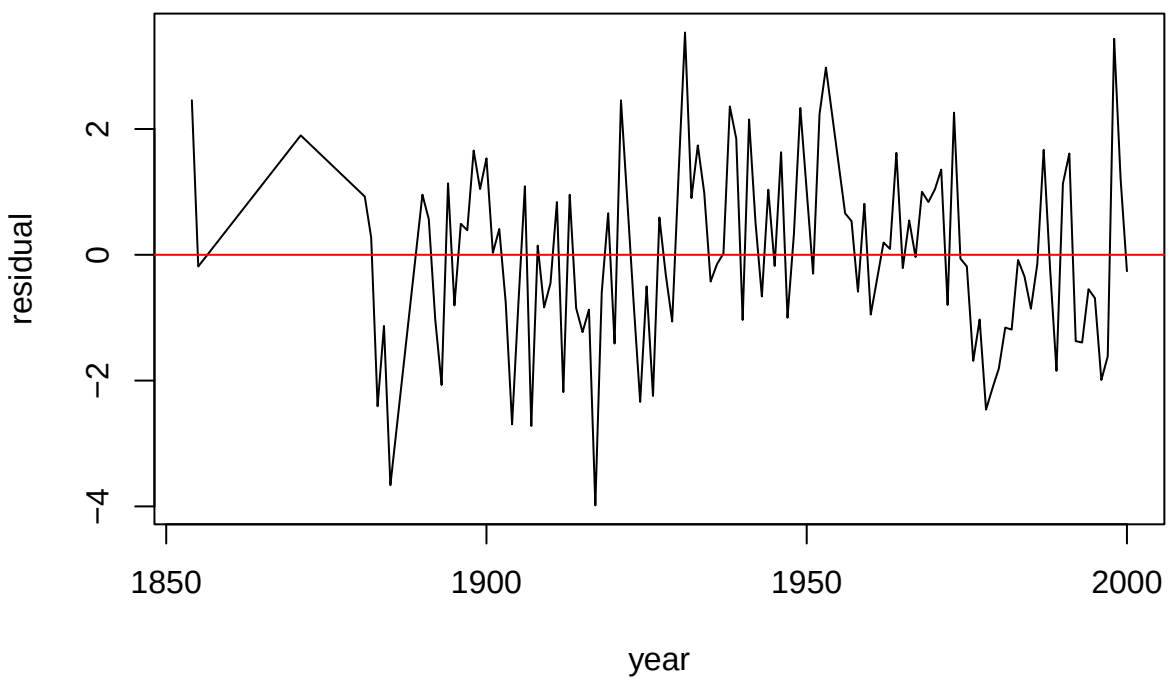
ii. Observations in successive years may be correlated. Fit a model that estimates this correlation. Does this change your opinion about the trend?

Fit a model that estimates this correlation.

First plot the residuals against year.

```
e = fit1$residuals
```

```
plot(e~dat1$year, type = "l", xlab = "year", ylab = "residual")
abline(h=0, col = "red")
```



It seems that overall the residuals tend to be large when the previous ones are large and tend to be small otherwise. Such result motivates me to plot the pairwise scatter plots for the residuals with different lags so as to make rudimentary guess towards the autocorrelation.

```
e = e[77:115] # only use data lagged by one
cor.data = as.data.frame(cbind(e[-c(58:60)], e[-c(1,59,160)],
            e[-c(1,2,60)], e[-c(1,2,3)]))
```

```
## Warning in cbind(e[-c(58:60)], e[-c(1, 59, 160)], e[-c(1, 2, 60)], e[-c(1, :
## number of rows of result is not a multiple of vector length (arg 2)
```

```
colnames(cor.data) = c("raw residual","lag1 residual","lag2 residual","lag3 residual")
pairs.panels(cor.data,
             hist.col = "#00AFBB",
             density = TRUE,  # show density plots
             ellipses = TRUE, # show correlation ellipses
```

```
            cex.cor = 0.6,
            stars = TRUE,
            pch = 19)
```

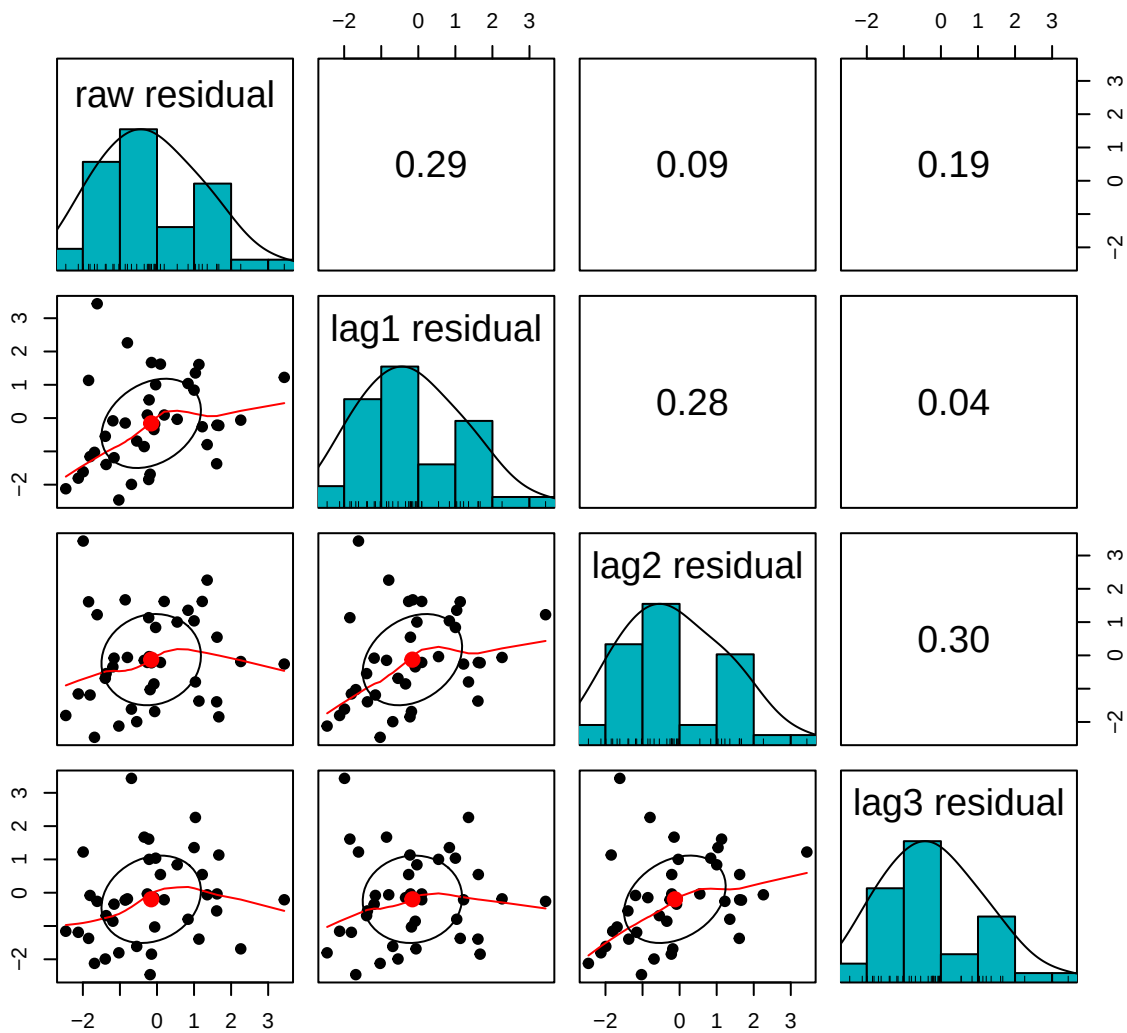


Figure 2: Pairwise scatter plots for raw residual, lag2 residual, lag3 residual and lag3 residual

Figure 2 shows the pairwise scatter plots just mentioned with the autocorrelations of different number of lags shown at the upper right. It suggests that lag 1 and lag3 autocorrelation might be the most dominated. However, since in practice it is usually preferred to choose the number of lags as small as possible, AR1 model is chosen when accounting for the autocorrelation in the following analysis.

Let l denote the number of lags. I then consider the following model. The model is fitted with generized least square estimator. The fitted result is shown in Table 2.

$$temp_t = \beta_0 + \beta_1\, t + \epsilon_t,\ E(\epsilon_t) = 0,\ Cov(\epsilon_t, \epsilon_{t-l}) = \sigma^2 \rho^l$$

```
g1 = gls(temp~year, data = dat1, correlation = corAR1(form = ~year))
summ = summary(g1)
summ.kable(summ$tTable)
```

Table 2: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	25.1840726	8.9718637	2.807006	0.0058912
year	0.0116403	0.0046265	2.516015	0.0132734

```
CI = intervals(g1)
CI$corStruct
```

```
##           lower      est.     upper
## Phi1 0.02920118 0.2303887 0.4136364
## attr(,"label")
## [1] "Correlation structure:"
```

From the output above, we can see that $\hat{\rho} = 0.2303887$ with 95% confidence interval $[0.02920118, 0.4136364]$. Since the confidence interval does not contains 0, I conclude that the lag 1 autocorrelation *rho* should be positive and significant.

Does this change your opinion about the trend?

Table 2 shows that year is still a significant predictor of temperature. Besides, its coefficient is positive, which has nothing contradicting with that shown if Table 1. Hence, the result so far does not change my opinion about the trend.

iii. Fit a polynomial model with degree 10 and use backward elimination to reduce the degree of the model. Plot your fitted model on top of the data. Use this model to predict the temperature in 2020.

Model

$$temp = \beta_0 + \beta_1\, year + \beta_2\, year^2 + \beta_3\, year^3 + \beta_4\, year^4 + \beta_5\, year^5 + \beta_6\, year^6 + \beta_7\, year^7 + \beta_8\, year^8 + \beta_9\, year^9 + \beta_{10}\, year^{10} + \epsilon$$

```
fit = lm(temp~poly(year,10), data = dat1); summary(fit)
```

```
##
## Call:
## lm(formula = temp ~ poly(year, 10), data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4987 -0.8641 -0.1745  1.1450  3.4255
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       47.7426     0.1319 361.927  < 2e-16 ***
## poly(year, 10)1    4.7616     1.4146   3.366  0.00107 **
```

```
## poly(year, 10)2   -0.9071     1.4146  -0.641  0.52277
## poly(year, 10)3   -3.3132     1.4146  -2.342  0.02108 *
## poly(year, 10)4    2.4383     1.4146   1.724  0.08774 .
## poly(year, 10)5    3.3824     1.4146   2.391  0.01860 *
## poly(year, 10)6    1.2124     1.4146   0.857  0.39337
## poly(year, 10)7   -0.9373     1.4146  -0.663  0.50908
## poly(year, 10)8   -1.1011     1.4146  -0.778  0.43812
## poly(year, 10)9    1.3994     1.4146   0.989  0.32483
## poly(year, 10)10   0.3474     1.4146   0.246  0.80652
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.415 on 104 degrees of freedom
## Multiple R-squared:  0.2165, Adjusted R-squared:  0.1411
## F-statistic: 2.873 on 10 and 104 DF,  p-value: 0.003335
```

Backward elimination to reduce the degree of the model

At each step, I reduced the degree whenever the p-value of the highest-degree term is over 0.05. The procedure only depends on the highest-degree term at each step so as to follow hierarchical principle.

```r
fit = lm(temp~poly(year,9), data = dat1); summary(fit)
```

```
##
## Call:
## lm(formula = temp ~ poly(year, 9), data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4836 -0.8831 -0.2156  1.1354  3.3936
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     47.7426     0.1313 363.557  < 2e-16 ***
## poly(year, 9)1   4.7616     1.4083   3.381  0.00101 **
## poly(year, 9)2  -0.9071     1.4083  -0.644  0.52089
## poly(year, 9)3  -3.3132     1.4083  -2.353  0.02050 *
## poly(year, 9)4   2.4383     1.4083   1.731  0.08631 .
## poly(year, 9)5   3.3824     1.4083   2.402  0.01807 *
## poly(year, 9)6   1.2124     1.4083   0.861  0.39123
## poly(year, 9)7  -0.9373     1.4083  -0.666  0.50716
## poly(year, 9)8  -1.1011     1.4083  -0.782  0.43605
## poly(year, 9)9   1.3994     1.4083   0.994  0.32265
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.408 on 105 degrees of freedom
## Multiple R-squared:  0.216,  Adjusted R-squared:  0.1488
## F-statistic: 3.215 on 9 and 105 DF,  p-value: 0.001769
```

```r
fit = lm(temp~poly(year,8), data = dat1); summary(fit)
```

```
##
```

```
## Call:
## lm(formula = temp ~ poly(year, 8), data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6086 -0.8600 -0.2385  1.0608  3.3975
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     47.7426     0.1313 363.579  < 2e-16 ***
## poly(year, 8)1   4.7616     1.4082   3.381  0.00101 **
## poly(year, 8)2  -0.9071     1.4082  -0.644  0.52085
## poly(year, 8)3  -3.3132     1.4082  -2.353  0.02047 *
## poly(year, 8)4   2.4383     1.4082   1.732  0.08626 .
## poly(year, 8)5   3.3824     1.4082   2.402  0.01805 *
## poly(year, 8)6   1.2124     1.4082   0.861  0.39118
## poly(year, 8)7  -0.9373     1.4082  -0.666  0.50713
## poly(year, 8)8  -1.1011     1.4082  -0.782  0.43600
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.408 on 106 degrees of freedom
## Multiple R-squared:  0.2086, Adjusted R-squared:  0.1489
## F-statistic: 3.494 on 8 and 106 DF,  p-value: 0.001284
```

```r
fit = lm(temp~poly(year,7), data = dat1); summary(fit)
```

```
##
## Call:
## lm(formula = temp ~ poly(year, 7), data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5922 -0.9032 -0.2322  0.9880  3.2941
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     47.7426     0.1311 364.241  < 2e-16 ***
## poly(year, 7)1   4.7616     1.4056   3.388 0.000988 ***
## poly(year, 7)2  -0.9071     1.4056  -0.645 0.520083
## poly(year, 7)3  -3.3132     1.4056  -2.357 0.020234 *
## poly(year, 7)4   2.4383     1.4056   1.735 0.085672 .
## poly(year, 7)5   3.3824     1.4056   2.406 0.017828 *
## poly(year, 7)6   1.2124     1.4056   0.863 0.390303
## poly(year, 7)7  -0.9373     1.4056  -0.667 0.506341
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.406 on 107 degrees of freedom
## Multiple R-squared:  0.2041, Adjusted R-squared:  0.152
## F-statistic: 3.919 on 7 and 107 DF,  p-value: 0.0007651
```

```r
fit = lm(temp~poly(year,6), data = dat1); summary(fit)
```

```
##
## Call:
## lm(formula = temp ~ poly(year, 6), data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6846 -0.8825 -0.1428  0.9388  3.2950
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     47.7426     0.1307 365.181  < 2e-16 ***
## poly(year, 6)1   4.7616     1.4020   3.396 0.000957 ***
## poly(year, 6)2  -0.9071     1.4020  -0.647 0.518996
## poly(year, 6)3  -3.3132     1.4020  -2.363 0.019905 *
## poly(year, 6)4   2.4383     1.4020   1.739 0.084851 .
## poly(year, 6)5   3.3824     1.4020   2.413 0.017527 *
## poly(year, 6)6   1.2124     1.4020   0.865 0.389067
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.402 on 108 degrees of freedom
## Multiple R-squared:  0.2008, Adjusted R-squared:  0.1564
## F-statistic: 4.522 on 6 and 108 DF,  p-value: 0.0003978
```

```r
fit2 = lm(temp~poly(year,5), data = dat1); summ.kable(summary(fit2)$coef)
```

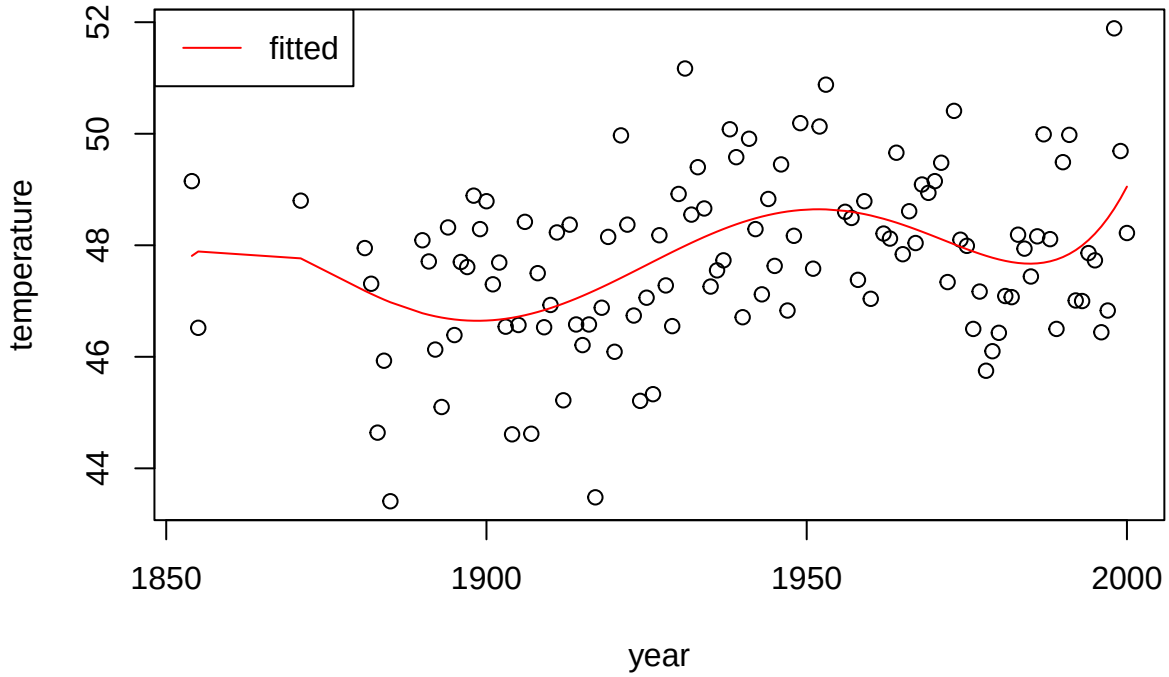
Table 3: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	47.7426087	0.1305855	365.6042581	0.0000000
poly(year, 5)1	4.7615720	1.4003733	3.4002162	0.0009418
poly(year, 5)2	-0.9071098	1.4003733	-0.6477628	0.5185004
poly(year, 5)3	-3.3132429	1.4003733	-2.3659712	0.0197491
poly(year, 5)4	2.4383320	1.4003733	1.7412014	0.0844698
poly(year, 5)5	3.3823682	1.4003733	2.4153333	0.0173841

We stopped the procedure at degree = 5 since the p-value of the term with degree 5 is now smaller than 0.05.

Plot your fitted model on top of the data.

```r
plot(temp~year, ylab = "temperature", data = dat1)
lines(dat1$year, fitted(fit2), col = "red")
legend("topleft", "fitted", lty = 1, col = "red")
```



The fit seems to be quite satisfactory.

Use this model to predict the temperature in 2020.

```
predict(fit2, newdata = data.frame(year = 2020), interval = "prediction")
```

```
##        fit      lwr      upr
## 1 60.07774 49.84092 70.31456
```

It predicts that the average temperature will be $60.07774^{\circ}F$ in 2020 with 95% confidence interval $[49.84092, 70.31456]$. This predicted value is not sensible since it appears too high. Such unusual result might be caused by the problem of extrapolation.

iv. Suppose someone claims that the temperature was constant until 1930 and then began a linear trend. Fit a model corresponding to this claim. What does the fitted model say about this claim?

Fit a model corresponding to this claim

Model

$$temp_t = \beta_0 + \beta_1 (t - 1930)_+ + \epsilon_t$$

```
temp = dat1$temp
year.pos = dat1$year-1930
year.pos[year.pos<=0] = 0
fit3 = lm(temp~year.pos)
```

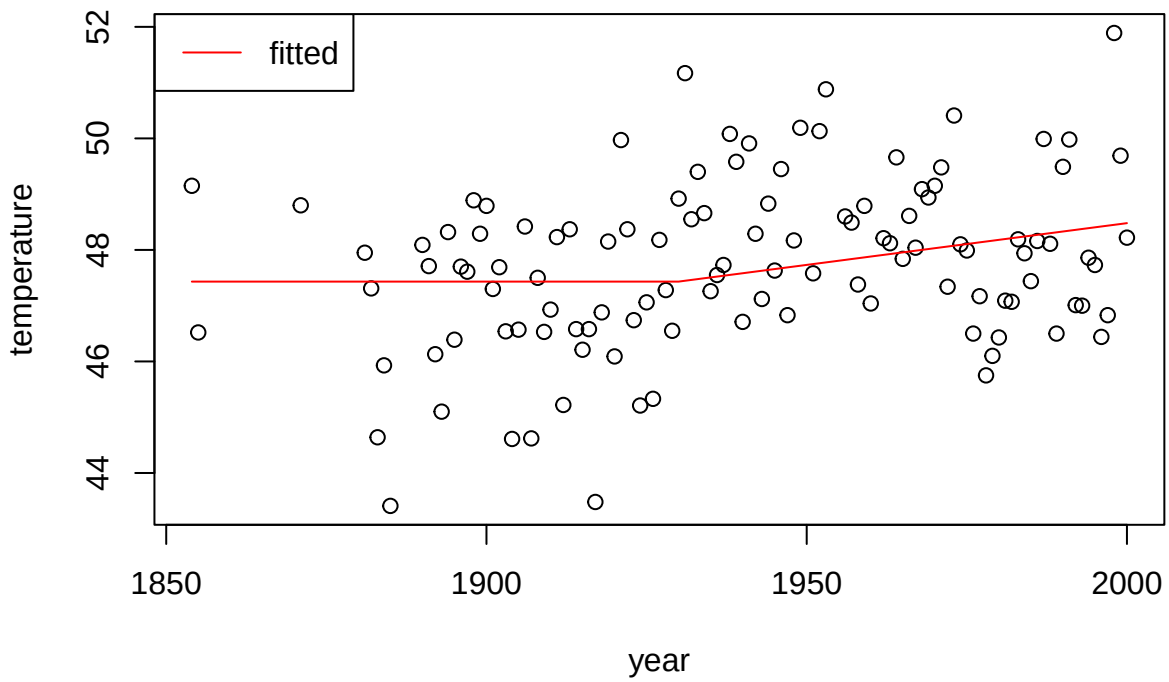
```
summ = summary(fit3)$coef
rownames(summ)[2] = "$(t-1930)_{+}$"
summ.kable(summ)
```

Table 4: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	47.4321507	0.1846044	256.939413	0.0000000
$(t-1930)_+$	0.0149697	0.0058573	2.555742	0.0119242

Plot your fitted model on top of the data.

```
plot(temp~year, ylab = "temperature", data = dat1)
lines(dat1$year, fitted(fit3), col = "red")
legend("topleft", "fitted", lty = 1, col = "red")
```



What does the fitted model say about this claim?

I compared the current model with the previous ones according to their root mean squared error (RMSE).

```
RMSE = matrix(c(summary(fit1)$sigma, summary(g1)$sigma,
              summary(fit2)$sigma), nrow = 1)
colnames(RMSE) = c("(i)", "(ii)", "(iii)"); rownames(RMSE) = "RMSE"
RMSE
```

```
##           (i)     (ii)    (iii)
## RMSE 1.466258 1.475718 1.400373
```

Table 4 shows that the coefficient for $(year - 1930)_+$ is positive and significant at 0.05 significance level. Besides, from the output above we can see that the RMSE of this model is the lowest so far. Hence I conclude that the claim might be correct.

v. Make a cubic spline fit with 6 basis functions evenly spaced on the range. Plot the fit in comparison to the previous fits. Does this model fit better than the straight line model?

Make a cubic spline fit with 6 basis functions evenly spaced on the range.

Model

```
(max(dat1$year) - min(dat1$year))/3
```

```
## [1] 48.66667
```

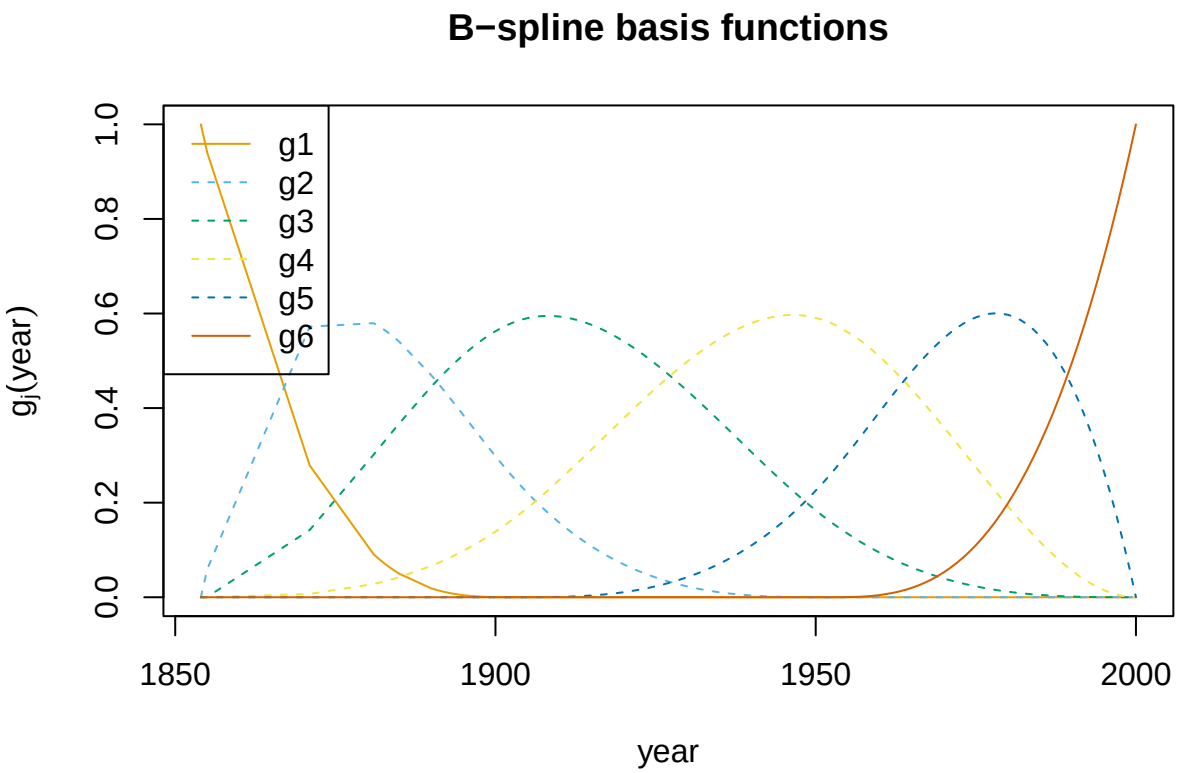
```
year.1 = min(dat1$year)+49; year.2 = min(dat1$year)+98;
c(year.1, year.2)
```

```
## [1] 1903 1952
```

$$temp_t = \sum_{j=1}^{6} g_j(t) + \epsilon_t,$$

where $g_1, g_2, \dots, g_6$ are shown as follows.

```
knots = c(1854,1854,1854,1854,1903,1952,2000,2000,2000,2000)
byear = splineDesign(knots, dat1$year)
matplot(dat1$year, byear,type="l", xlab = "year", ylab = expression(g[j](year)),
        main="B-spline basis functions", col = cbp[2:7], lty = c(1,2,2,2,2,1))
legend("topleft", paste("g", 1:6, sep = ""), col = cbp[2:7], lty = c(1,2,2,2,2,1))
```




```
fit4 = lm(dat1$temp~byear-1)
```

Table **??** shows the fitted result.

```
summ = summary(fit4)$coef
rownames(summ) = paste("g", 1:6, sep = "")
summ.kable(summ)
```

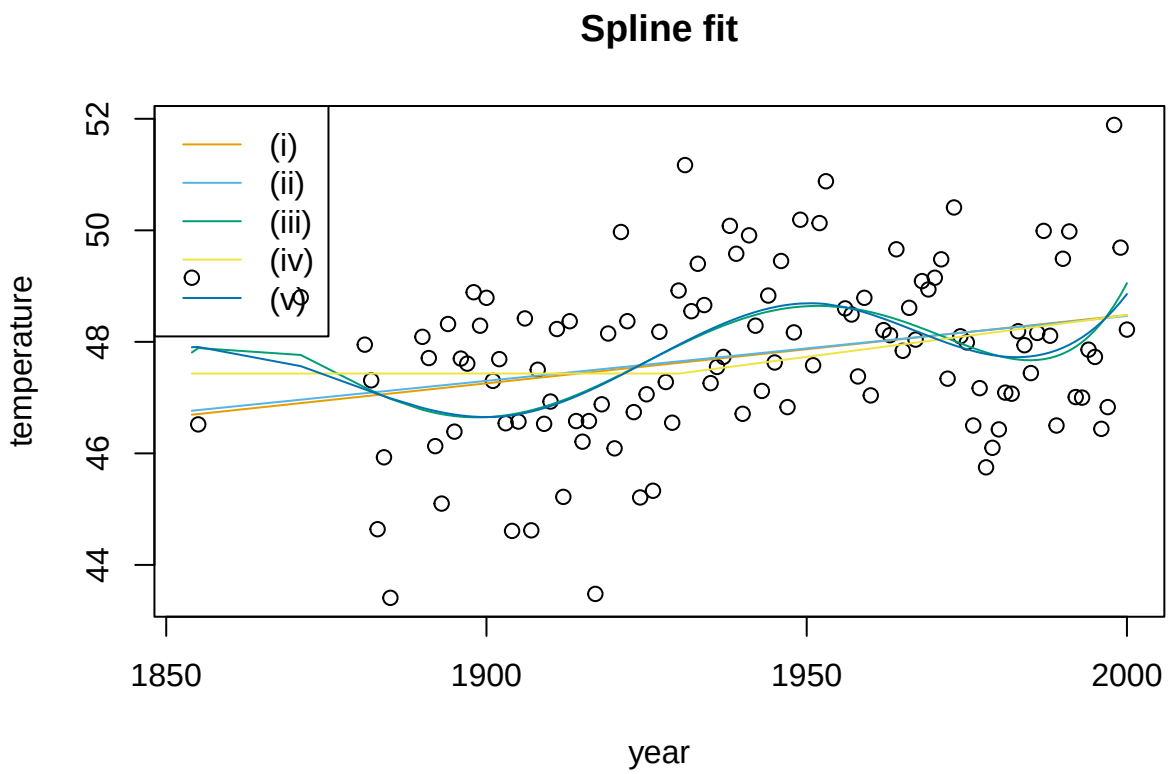
Table 5: Summary for the fitted result

	Estimate	SE	t-value	p-value
g1	47.90404	1.0136595	47.25851	0
g2	48.01678	1.1612896	41.34781	0
g3	44.92541	0.9157573	49.05820	0
g4	50.70812	0.8498659	59.66602	0
g5	46.47736	0.7423626	62.60736	0
g6	48.85721	0.6285363	77.73173	0

Plot the fit in comparison to the previous fits.

```
plot(dat1$year, dat1$temp, xlab = "year", ylab="temperature", lty=1, main="Spline fit")
lines(dat1$year, fit1$fitted, col = cbp[2])
lines(dat1$year, g1$fitted, col = cbp[3])
```

```
lines(dat1$year, fit2$fitted, col = cbp[4])
lines(dat1$year, fit3$fitted, col = cbp[5])
lines(dat1$year, fit4$fitted, col = cbp[6])
legend("topleft", c("(i)", "(ii)", "(iii)", "(iv)", "(v)"), col = cbp[2:6], lty = 1)
```



Does this model fit better than the straight line model?

I compared the current model with the previous ones according to their root mean squared error (RMSE).

```
RMSE = matrix(c(summary(fit1)$sigma, summary(g1)$sigma,
              summary(fit2)$sigma, summary(fit3)$sigma, summary(fit4)$sigma),
              nrow = 1)
colnames(RMSE) = c("(i)", "(ii)", "(iii)", "(iv)", "(v)"); rownames(RMSE) = "RMSE"
RMSE
```

```
##            (i)      (ii)     (iii)      (iv)       (v)
## RMSE 1.466258 1.475718 1.400373 1.490674 1.400822
```

Yes, I think this model fit is much better than the straight line model since the RMSE of it is smaller than that of (i) and (ii).

0.2 Q2 The data contains Infant Mortality Rates (IMR) and Physical Quality of Life Index (PQLI) scores, which is an indicator of average wealth, for selected Indian States. Using the data set, construct a single model for infant mortality rate, using suitably defined dummy variables for rural-urban and male-female distinctions. You should investigate whether there is a male-female and/or rural-urban difference in mortality rate after adjusting for other covariates.

```
dat2 = read.table("E1.20.txt")
colnames(dat2) = c("State", "PQLI.Score", "Combined IMR",
                   "Rural.Male.IMR", "Rural.Female.IMR",
                   "Urban.Male.IMR", "Urban.Female.IMR")
```

Convert the data into long format

```
dat2 = data.frame(State = factor(rep(dat2$State, each = 4)),
                  PQLI = rep(dat2$PQLI.Score, each = 4),
                  Gender = factor(rep(c("Male", "Female", "Male", "Female"),
                                      length(dat2$State))),
                  Region = factor(rep(c("Rural", "Rural", "Urban", "Urban"),
                                      length(dat2$State))),
                  IMR = as.vector(t(as.matrix(dat2[,4:7]))))
levels(dat2$Gender)
```

```
## [1] "Female" "Male"
```

```
levels(dat2$Region)
```

```
## [1] "Rural" "Urban"
```

Note the dataset is now consisting of three categorical variables, including State, Gender, Region, and two quantitative variables, PQLI and IMR.

Model

For the two categorical preditors, Gender and Region, Female and Rural are set respectively to be the baselines for the former and the later. I first consider a full model as follows.

$$\begin{aligned} IMR =& \beta_0 + \beta_1\, PQLI + \beta_2\, Gender + \beta_3\, Region + \beta_{12}\, PQLI \times Gender + \\ & \beta_{13}\, PQLI \times Region + \beta_{23}\, Gender \times Region + \beta 123\, PQLI \times Gender \times Region + \epsilon, \end{aligned}$$

where *Gender* and *Region* are dummy variables defined as follows.

$$Gender = \begin{cases} 1, \text{ if Gender is Male} \\ 0, \text{ o.w.} \end{cases}$$

and

$$Region = \begin{cases} 1, \text{ if Region} = \text{Urban} \\ 0, \text{ o.w.} \end{cases}$$

Backward elimination and summary for the fitted result

In the following, Backward elimination procedure is adopted to remove nonsignificant predictors in the full model. At each stage, the predictor with the largest p-values were removed while following hierarchical principle.

```
fit = lm(IMR~PQLI*Gender*Region, data = dat2); summ.kable(summary(fit)$coef)
```

Table 6: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	181.4581246	10.3419247	17.5458757	0.0000000
PQLI	-1.5493916	0.2167853	-7.1471231	0.0000000
GenderMale	-2.4328530	14.6256901	-0.1663411	0.8686506
RegionUrban	-77.9536585	14.6256901	-5.3299132	0.0000032
PQLI:GenderMale	0.1583517	0.3065808	0.5165091	0.6080853
PQLI:RegionUrban	0.7798543	0.3065808	2.5437158	0.0145567
GenderMale:RegionUrban	10.5944726	20.6838493	0.5122099	0.6110657
PQLI:GenderMale:RegionUrban	-0.3740780	0.4335707	-0.8627843	0.3929345

```
fit = update(fit, .~.-PQLI:Gender:Region); summ.kable(summary(fit)$coef)
```

Table 7: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	177.3864295	9.1762232	19.3310937	0.0000000
PQLI	-1.4558721	0.1872077	-7.7767760	0.0000000
GenderMale	5.7105372	11.1407587	0.5125806	0.6107517
RegionUrban	-69.8102683	11.1407587	-6.2662042	0.0000001
PQLI:GenderMale	-0.0286872	0.2161688	-0.1327076	0.8950163
PQLI:RegionUrban	0.5928153	0.2161688	2.7423725	0.0087234
GenderMale:RegionUrban	-5.6923077	8.4305664	-0.6751987	0.5030043

```
fit = update(fit, .~.-PQLI:Gender); summ.kable(summary(fit)$coef)
```

Table 8: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	178.0109288	7.7931328	22.8420243	0.0000000
PQLI	-1.4702157	0.1512134	-9.7227865	0.0000000
GenderMale	4.4615385	5.8973114	0.7565377	0.4531843
RegionUrban	-69.8102683	11.0211541	-6.3342067	0.0000001
PQLI:RegionUrban	0.5928153	0.2138481	2.7721335	0.0080129
GenderMale:RegionUrban	-5.6923077	8.3400578	-0.6825262	0.4983292

```
fit = update(fit, .~.-Gender:Region); summ.kable(summary(fit)$coef)
```

Table 9: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	179.4340057	7.4662445	24.032699	0.0000000
PQLI	-1.4702157	0.1503517	-9.778512	0.0000000
GenderMale	1.6153846	4.1462649	0.389600	0.6985922
RegionUrban	-72.6564221	10.1436610	-7.162742	0.0000000
PQLI:RegionUrban	0.5928153	0.2126294	2.788022	0.0076303

```
fit = update(fit, .~.-Gender); summ.kable(summary(fit)$coef)
```

Table 10: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	180.2416980	7.1089947	25.354035	0.0000000
PQLI	-1.4702157	0.1490173	-9.866073	0.0000000
RegionUrban	-72.6564221	10.0536368	-7.226880	0.0000000
PQLI:RegionUrban	0.5928153	0.2107423	2.812987	0.0070917

Table 10 shows the fitted result for the final model.

Investigate whether there is a male-female and/or rural-urban difference in mortality rate after adjusting for other covariates.

We can make comment regarding the male-female and rural-urban difference in mortality rate after adjusting for other covariates according to Table 10.

1. After adjusting for PQLI, it is found that IMR of the male might not differ from that of the female.
2. After adjusting for PQLI, IMR in Urban region seems to be significantly lower than that in the Rural region under significance level 0.05.

0.3 Q3 Data on the relationship between corn yield (bushels per acre) and nitrogen (pounds per acre) fertilizer application were studies in Wisconsin in 1994. Use transformations to find a good model for predicting yield from nitrogen. Use a goodness of fit test to check your model.

```
dat3 = read.table("cornnit.txt", header = TRUE)
table(dat3$nitrogen)
```

```
##
##   0  50  75 100 150 200 300
##  11   7   4   7   4   7   4
```

This is an experimental data, where nitrogen can be regarded as a factor consisting of 7 levels: 0, 50, 75, 100, 150, 200, 300.

For later goodness-of-fit test, we first fit the saturated model as follows.

$$\begin{aligned} yield = \beta_0 + \beta_1\, nitrogen_{50} + \beta_2\, nitrogen_{75} + \beta_3\, nitrogen_{100} \\ + \beta_4\, nitrogen_{150} + \beta_5\, nitrogen_{200} + \beta_6\, nitrogen_{300} + \epsilon, \end{aligned}$$

where $nitrogen_i$ is a dummy variable which values 1 if nitrogen is i pounds and values 0 otherwise. The fitted result for this saturated model is shown in Table 11, and the corresponding residual plot is shown in Figure 3.

```
full.fit = lm(yield~factor(nitrogen), data = dat3)
summ.kable(summary(full.fit)$coef)
```

Table 11: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	88.54545	4.484973	19.742694	0.00e+00
factor(nitrogen)50	44.02597	7.191957	6.121557	4.00e-07
factor(nitrogen)75	42.20455	8.685113	4.859412	2.18e-05
factor(nitrogen)100	51.31169	7.191957	7.134593	0.00e+00
factor(nitrogen)150	50.20455	8.685113	5.780529	1.20e-06
factor(nitrogen)200	56.45455	7.191957	7.849678	0.00e+00
factor(nitrogen)300	51.45455	8.685113	5.924453	8.00e-07

```
e = rstandard(full.fit)
plot(e~dat3$nitrogen, xlab = "nitrogen", ylab = "internally studentized residuals")
abline(h = 0, col = "red")
```

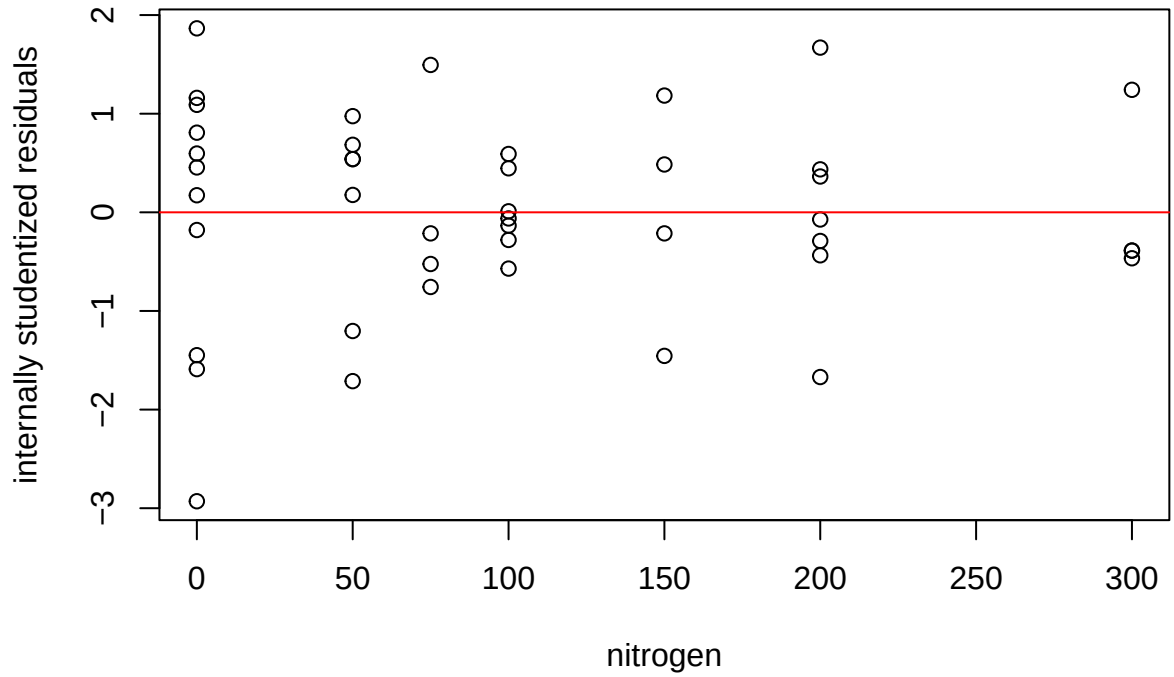


Figure 3: Internally studentized residual plot for the saturated model

Now, we fit a quite simple model as follows.

$$yield = \beta_0 + \beta_1\, nitrogen + \epsilon,$$

The fitted result is shown in Table 12.

```
fit = lm(yield~nitrogen, data = dat3)
summ.kable(summary(fit)$coef)
```

Table 12: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	107.4386437	4.6662152	23.024794	0.0e+00
nitrogen	0.1772966	0.0337716	5.249876	4.7e-06

To examine the goodness-of-fit of this current model by comparing it with the saturated model, we conduct ANOVA. The null and the alternative hypotheses for the corresponding F-test is as follows.

$$\begin{cases} H_0 : \text{the current model is adequate enough} \\ H_1 : \text{the current model is not adequate enough} \end{cases}$$

```r
anova(full.fit, fit)
```

```
## Analysis of Variance Table
##
## Model 1: yield ~ factor(nitrogen)
## Model 2: yield ~ nitrogen
##   Res.Df     RSS Df Sum of Sq      F    Pr(>F)
## 1     37  8186.8
## 2     42 17699.2 -5   -9512.4 8.5982 1.774e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The output above shows the ANOVA table, where we can see that the p-value is extremely small. Hence we reject H_0 and conclude that the current model might not be adequate enough.

```r
e = rstandard(fit)
plot(e~dat3$nitrogen, xlab = "nitrogen", ylab = "internally studentized residuals")
abline(h = 0, col = "red")
```

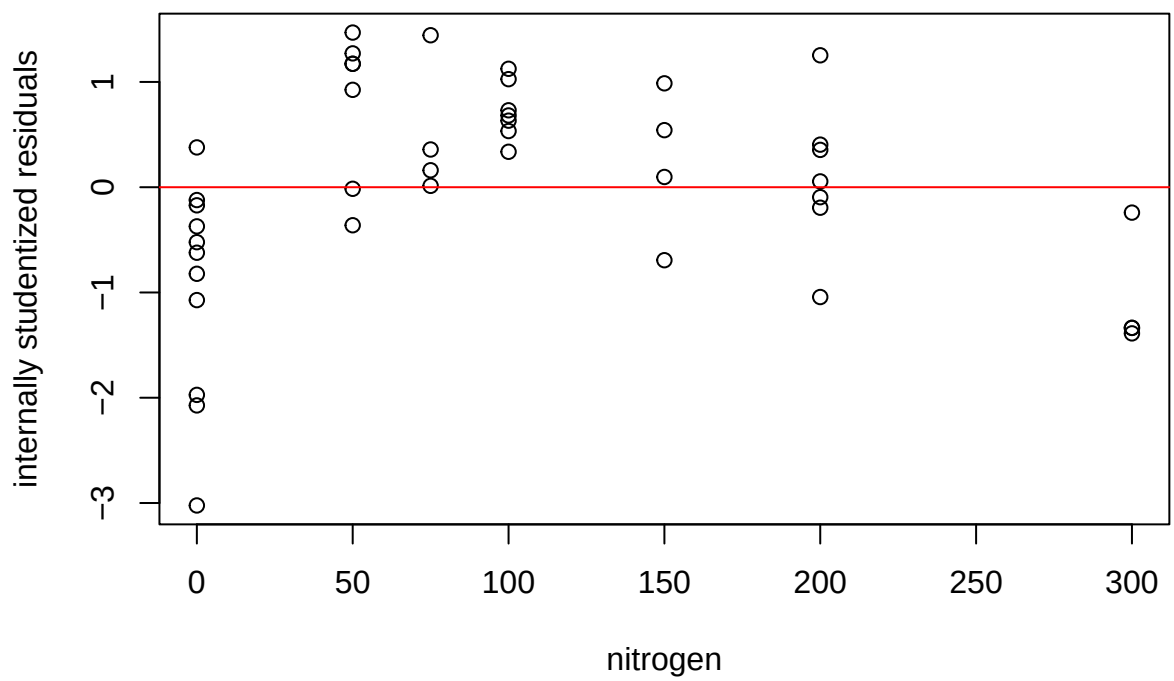


Figure 4: Internally studentized residual plot for the current model

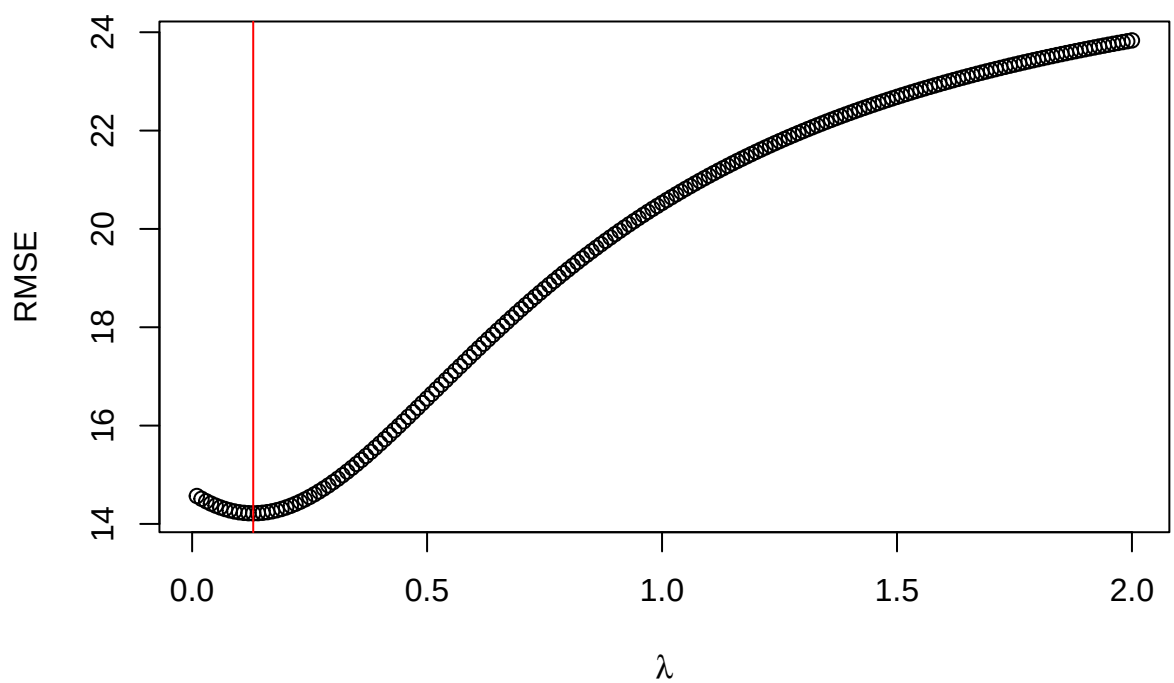
Figure 4 shows the internally studentized residual for the current model. It seems to suggest that some power transformation of *nitrogen* might be needed to improve the fitting.

To select a suitable λ for $nitrogen^{\lambda}$, I tried $\lambda \in \{0.01, 0.02, \ldots, 2\}$. For each value of λ, I fitted the model as follows.

$$yield = \beta_0 + \beta_1 \, nitrogen^{\lambda} + \epsilon.$$

Then I extracted and recorded the corresponding RMSE. After enumerating all those distinct values of λ, I chose the one that minimized RMSE as the suitable λ value for power transformation on *nitrogen*.

```
lambda = seq(0.01,2,0.01)
RMSE = rep(NA, length(lambda))
for(i in 1:length(lambda)){
  l = lambda[i]
  fit = lm(yield~I(nitrogen^l), data = dat3)
  RMSE[i] = summary(fit)$sigma
}
plot(lambda, RMSE, xlab = expression(lambda))
opt.lambda = lambda[which.min(RMSE)]
abline(v = opt.lambda, col = "red")
```




```
opt.lambda
```

```
## [1] 0.13
```

From the output above, it was found that $\lambda = 0.13$ seems to be the most suitable model since it leads to the lowest RMSE. Therefore, the following model is refitted.

$$yield = \beta_0 + \beta_1\, nitrogen^{0.13} + \epsilon.$$

The fitted result for this model is shown in Table 13, and the corresponding residual plot is shown in Figure 5.

```
fit = lm(yield~I(nitrogen^0.13), data = dat3)
summ = summary(fit)$coef
rownames(summ)[2] = "$\\text{nitrogen}^{0.13}$"
summ.kable(summ)
```

Table 13: Summary for the fitted result

	Estimate	SE	t-value	p-value
(Intercept)	88.53937	4.248174	20.84174	0
$\text{nitrogen}^{0.13}$	26.68282	2.628579	10.15104	0

```
e = rstandard(fit)
plot(e~dat3$nitrogen, xlab = "nitrogen", ylab = "internally studentized residuals")
abline(h = 0, col = "red")
```

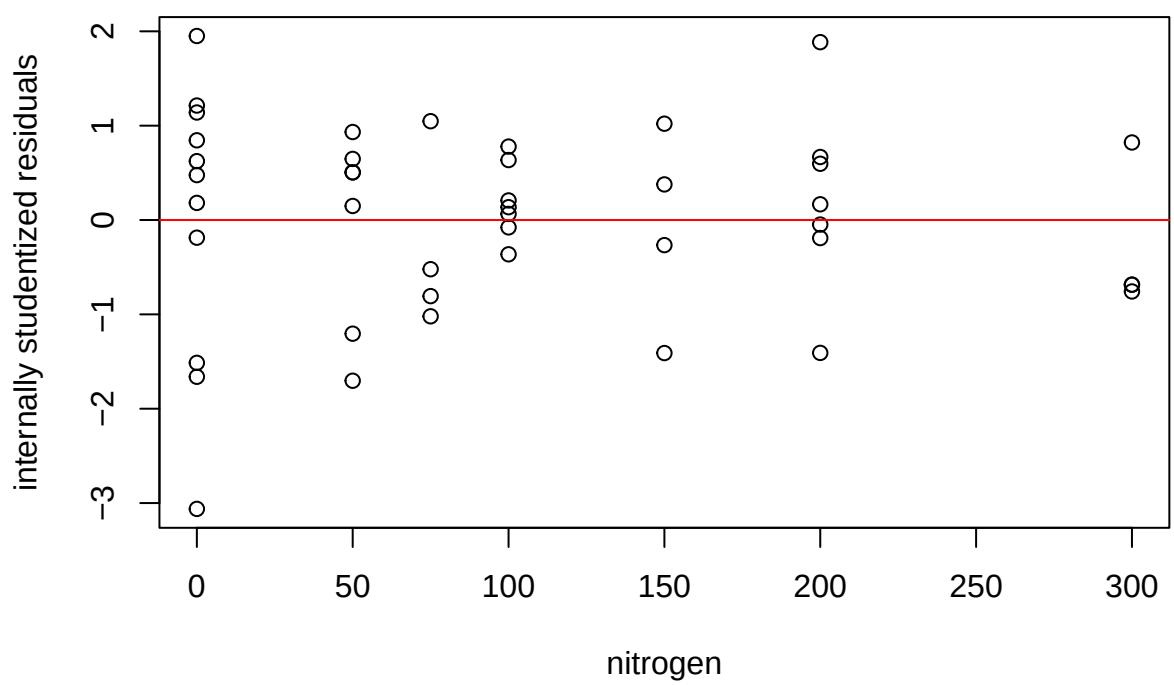


Figure 5: Internally studentized residual plot for the current model

Again, we conduct the ANOVA as that described before to examine the goodness-of-fit of the current model.

```
anova(full.fit, fit)
```

```
## Analysis of Variance Table
##
## Model 1: yield ~ factor(nitrogen)
## Model 2: yield ~ I(nitrogen^0.13)
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     37 8186.8
## 2     42 8488.3 -5   -301.52 0.2725 0.9252
```

The output above shows the ANOVA table, where we can see that the p-value is now greater than 0.05. Hence we do not reject H_0 and conclude that the current model might be adequate enough.